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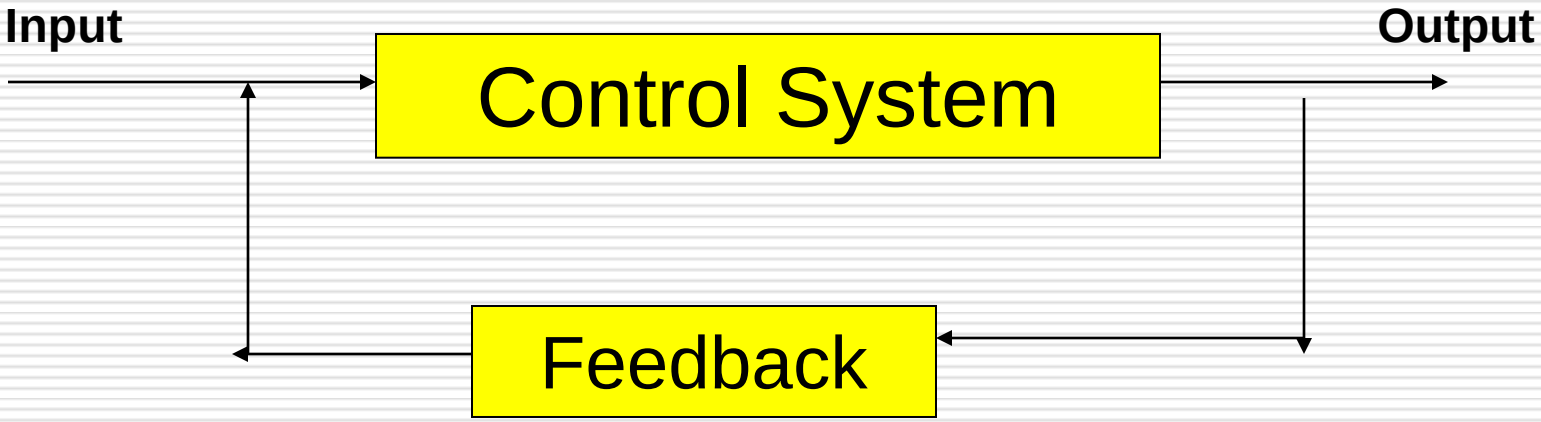
# Controls and BAS

- Whats controls ?
- Why do we need them?

# Control system

- Definition

# Control schematic



# What's controls ?

- The intelligence behind the mechanical systems
- Its like the human mind which processes millions of inputs every minute
- Human brain is the oldest and best control system

# History of controls ?

- Electrical controls- 1970’s
- Electronic discrete controls- 1980’s
- Digital controls -1990’s
- Open protocol controls- 2000

# Why BMS ?

- To provide one point monitoring and control of all utilities and integration with Fire life safety systems to offer precision control, automation, safety lockouts, energy savings and better information systems

# Benefits

- Precision control
- Automatic operations
- Safety Interlock
- Energy savings
- Better Maintenance management
- Good Information Systems

# Sub-systems of BAS

- Energy management systems
  - HVAC
  - Electrical
  - Lighting
  - Misc utilities like hydrant, water systems
- Fire life safety systems
  - Fire detection systems
  - Access control systems
  - CCTV systems
  - Perimeter protection systems

# Features

- HVAC
- Monitor and control high side equipment including chiller through integration on Modbus/Bacnet for trending of parameters
- Monitoring and control of Pumps and other associated equipment for High side
- Monitoring and control of RA temp , control of AHU control valve, on/off operations in time schedule programs,monitor filter clean/dirty conditions, control of VFD if provided,fire status integration, fire damper control if motorized and monitoring the IAQ parameters for each level in 2 locations.
- Monitoring of CO levels in basement and control of fans in the basement for ventilation and exhaust.
- Integration of precision air conditioner for server room/switch/hub/IDF rooms

# Features

- Electrical
- Monitor transformer temperatures and trips, breaker status ,monitor various parameters like V, A, PF, Freq for incomer and various important outgoing feeders
- Integration of DG PLC and monitoring critical engine and alternator parameters
- Integration of UPS for monitoring all critical parameters

# Features

- Lighting
- Control of lighting in general office areas based on time schedule and occupancy override facility with integration to the EPABX system.
- Control of lights externally based on lux levels , time schedule programs

# Features

- Plumbing, Sanitary & Fire fighting
- Monitoring and control of  levels in various tanks and control of pumps
- Time schedule control for landscape pumps
- Monitoring main pumps, jockey pumps, Engine driven pumps

# Proposed Design

- System Architecture based on TCP/IP with 2 layers of architecture – controllers and Head end based on Industry standard open protocol like ModBus/Bacnet
- System covering the following
  - HVAC
  - Lighting internal – Workstations/Architectural
  - Electrical utilities like power measurement, breakers
  - Indoor air quality
  - Misc Equipments like pumps , levels etc.

# Proposed design

- Integration with following equipment on Bacnet or Modbus
  - Fire detection Panels
  - DG’s
  - UPS
  - Breakers
  - VFD’s
  - Precision AC Units
  - Chillers

# Proposed design

- Maintenance Management software for pro-active maintenance and Facility Management
- Multi-user software with facility to monitor and operate from multiple workstations
- Hi-speed controllers with  freely programming capability
- BMS Network  integrated with facility wide LAN
- Web based capability using IP address for multi-location monitoring and control- for e.g from UK
- SMS Based alarming utility for messaging of key alarms to a mobile phone and facility to escalate

# Components of BAS

- Input devices
- Output devices
- Controllers
- Head end Workstation

# Input Devices

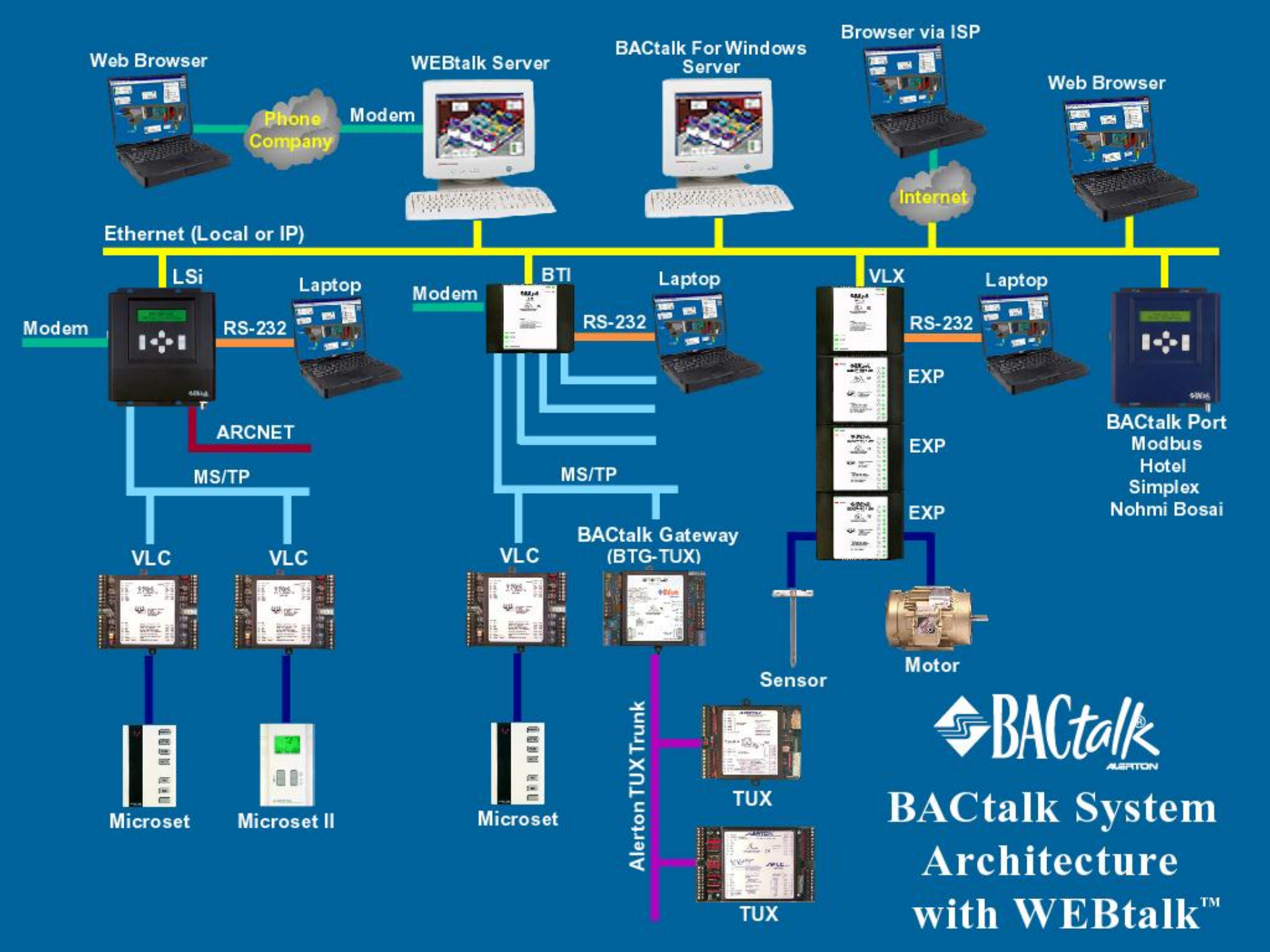
- Types
- Analog Input- Any input that varies with time for e.g. Temperature, humidity , Pressure
- Digital Input- Any input that works on a two state mode ( 0 or 1)like on or off , clean/dirty.
- Pulse input- Any input incremental in nature and totalizes over a period of time , KWH meter, water meter

# Controllers

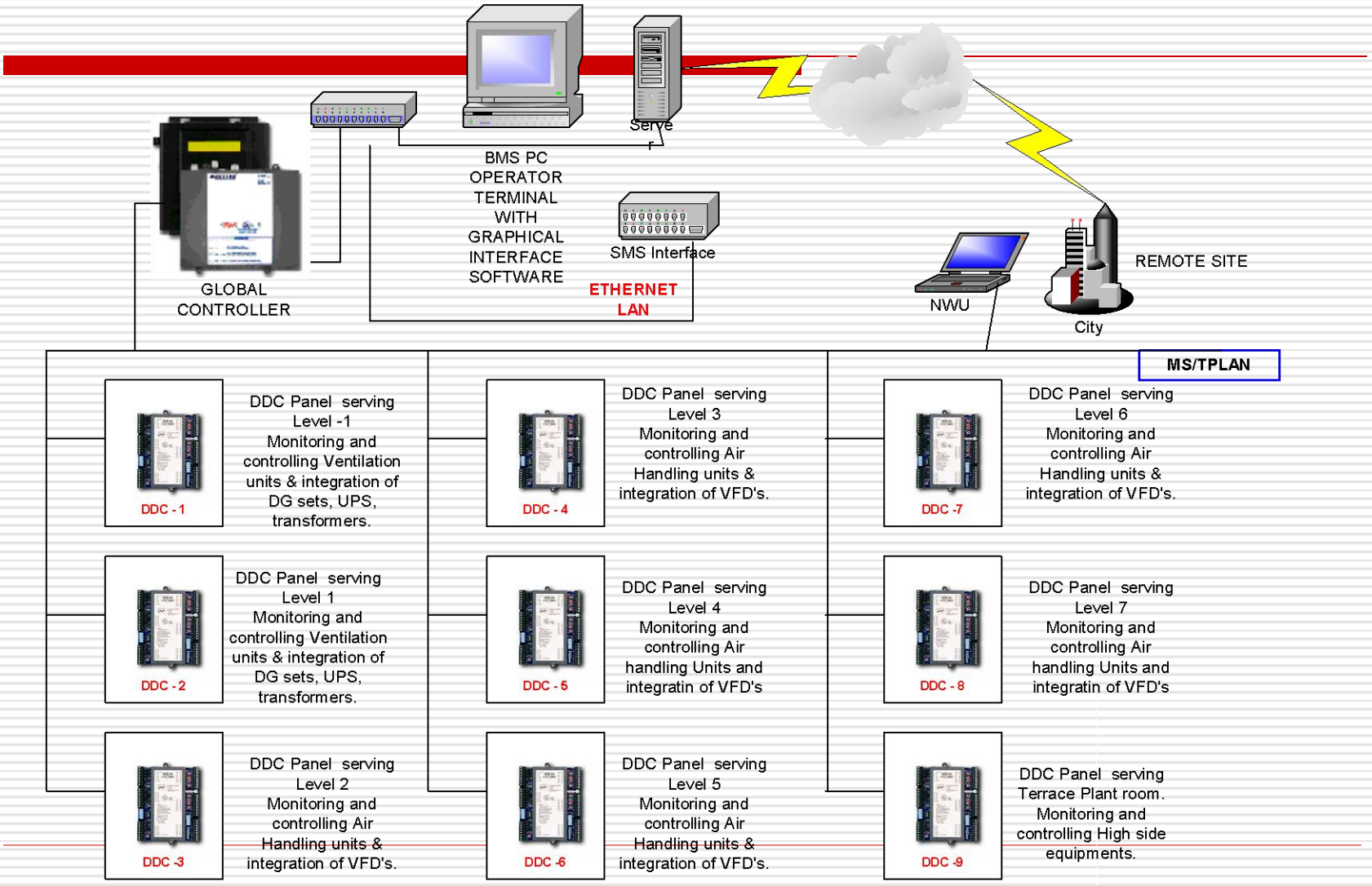
- Microprocessor based with real time clock and memory
- Programmable for various control logics and switching tables
- Capability to communicate with other controllers on communication protocol
- Stand alone operations

# Output devices

- Types
- Analog output- output that varies with time e.g 3 way valve , VFD
- Digital out put- on/off output e.g fan on/off , compressor on/off


Web Browser
Phone Company
Modem
WEBtalk Server
BACtalk For Windows Server
Browser via ISP
Internet
Web Browser
Ethernet (Local or IP)
LSi
Modem
RS-232
Laptop
ARCNET
MS/TP
VLC
VLC
Microset
Microset II
BTI
Modem
RS-232
Laptop
MS/TP
VLC
BACtalk Gateway (BTG-TUX)
Microset
Alerton TUX Trunk
TUX
TUX
VLX
RS-232
Laptop
EXP
EXP
EXP
Sensor
Motor
BACtalk Port
Modbus
Hotel
Simplex
Nohmi Bosai
BACtalk
ALERTON
BACtalk System Architecture with WEBtalk™

# System Architecture drawing



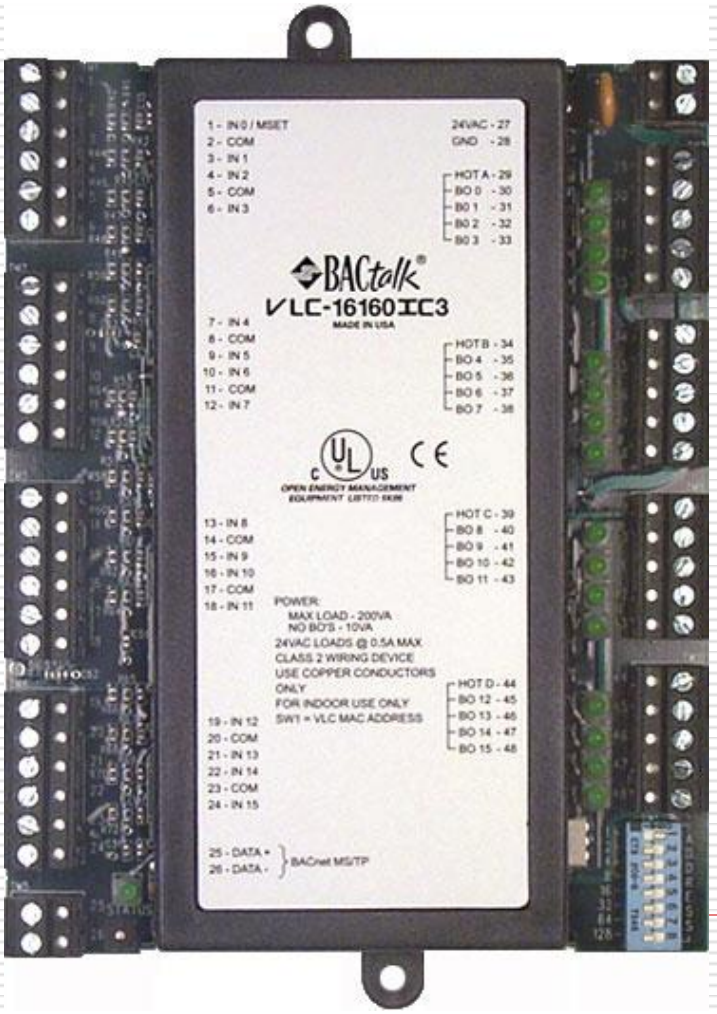
# BACtalk Visual Logic Controllers

- Capable 10-bit universal inputs
- Interoperable Fully BACnet-compliant on MS/TP LAN at up to 76.8 Kbps.
- Fast Internal logic loop of 100 msec.

# Visual Logic Controllers

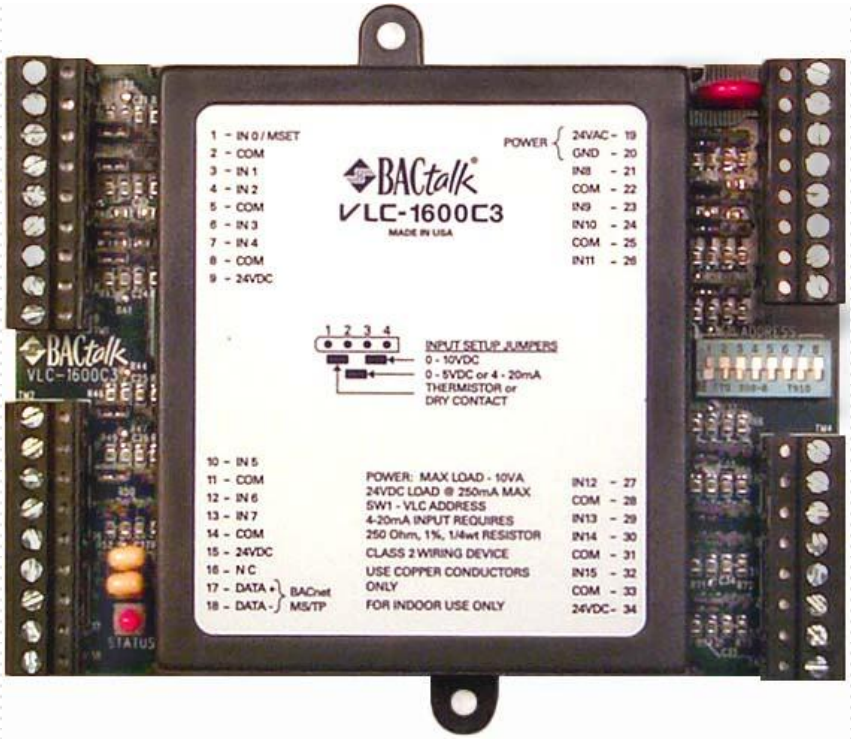
- Reliable Extensive onboard filtering, with all program data backed up in non-volatile EEPROM.
- Versatile Fully programmable for central plant systems, air handling units, clean rooms, fume hoods and other control and process equipment.

# VLC 16160IC3



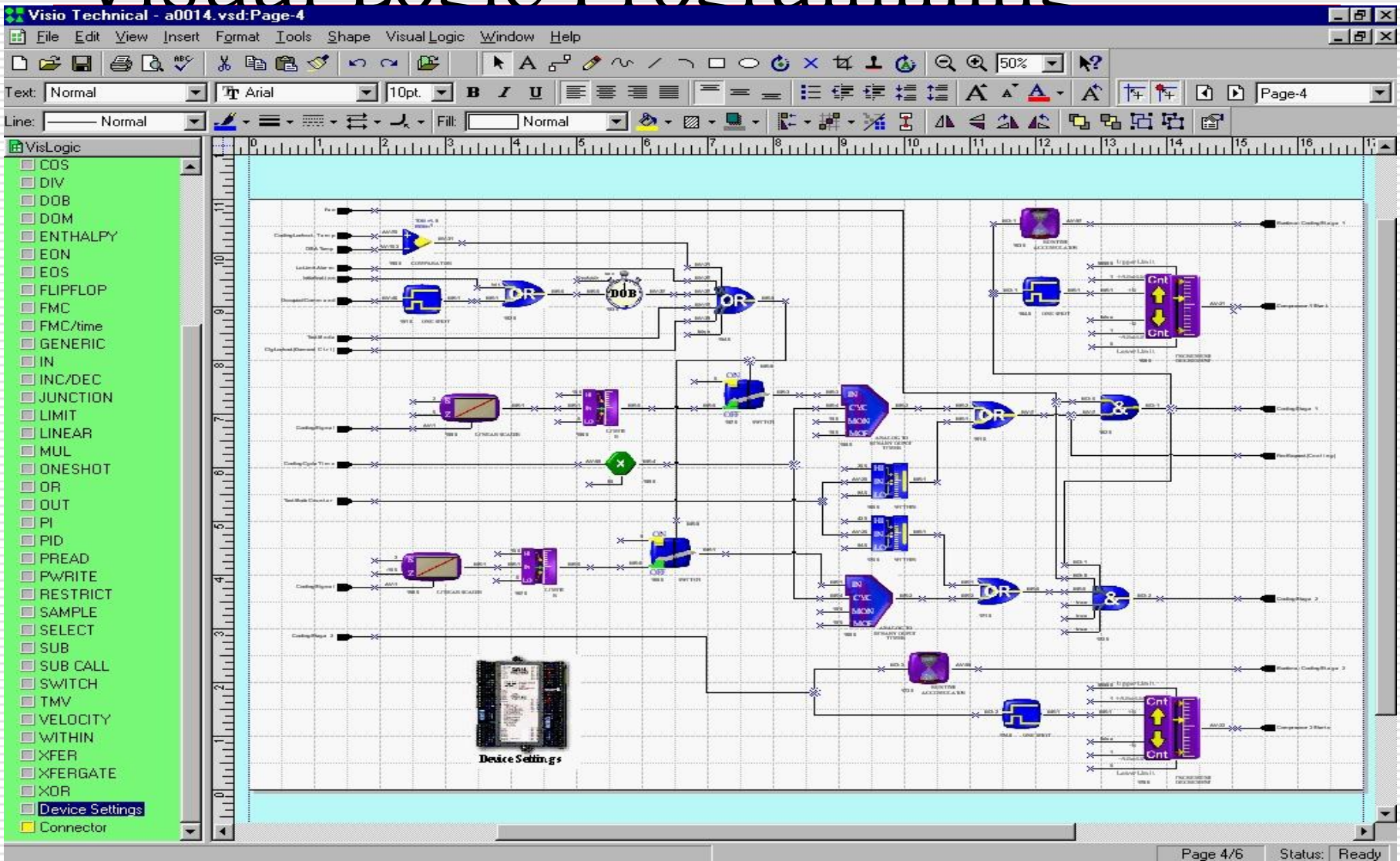

- 16 Universal Inputs
- 16 Binary Outputs grouped into 4 groups of four binary triac outputs, each with an isolated power source, making it ideal for control of packaged equipment

# VLC 1600C3




- 16 Universal Inputs

# Visual Logic Programming



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# Standards

- UL 817
- NFPA 72e
- IS2189
- UL 865
- UL 555

# What is Web enabled BAS

- A BAS which utilizes the internet to provide real-time access  to building personnel to monitor ,control with necessary web servers, front end software's and a static IP address
- Internet control goes beyond the simple access. Internet control automates facilities HVAC and processes while expanding the scope to enterprise level,thus converting BAS to MIS

# Internet and BAS

- A Web based presentation of dynamic building information allows not only operators to operate from anywhere with full functionality , it allows interaction with contractors, consultants, equipment suppliers to provide feed forward information to the building operation matrix
- It provides upper management with bottom line oriented screens to provide dynamic proof of the success or failure of building performance

# Internet based BAS

- What has helped is the standardization of data communications like TCP/IP
- It is the convergence of IT and BAS
- Web based training for operating staff
- Resource library for the building
- Multiple locations open system help in integration
- HTML for  dynamic data

# Advantages

- Standard web browser like IE or NN as a interface
- No proprietary software required
- High level of security
- No multiple copies of s/w required
- Unlimited users
- Companies with multi-locations facilities
- Unifies users , consultants , contractors to study the system performance and fine tune

# Building.com

- The concept of creating a website for each complex allows all critical building information to be located in one organized place for all concerned to access.
- The website is a focus point for energy performance,client comfort and complaints.

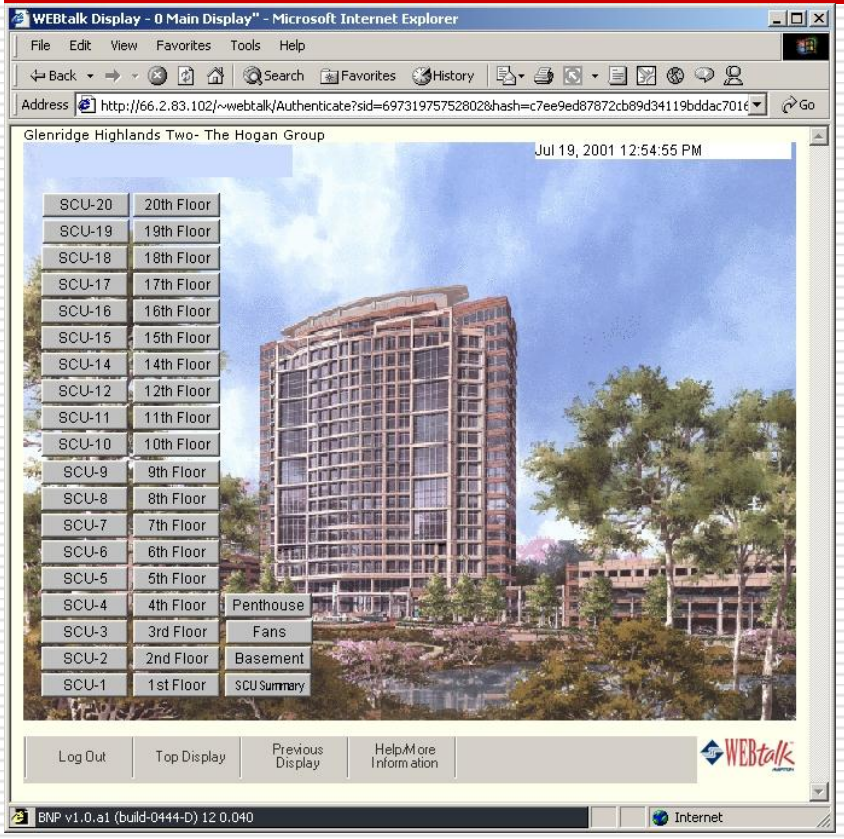
# Building.com

- Many large buildings have a .com address
- Very powerful and almost becoming a standard in US
- Real-time streaming video data to show construction progress
- Helps bidding process – all relevant drawings, documents for owners, bidders, consultants available on the web site
- Large buildings can sell internet access by leasing ISDN lines/ T1 connects and distributing to many users

# Building.com

- Buildings could sell  e-mail ID , server space
- It allows a transparent system for tenant billing systems
- Emergency breakdown communications, shutdowns, special security procedures
- Access to real time temp, IAQ data., virtual thermostats and lighting interfaces
- It is a building operations and maintenance tool

# WEBtalk




- Access your building using an Internet browser
- Increase building management efficiency
- Ease to use, install, maintain

# The End

- End of session 1
- Session 2- BMS sample design exercise